Linear algebra Recitation

Dimension of a Sum

If $\mathcal{X}$ and $\mathcal{Y}$ are subspaces of a vector space $\mathcal{V}$, then

$$\dim(\mathcal{X}+\mathcal{Y}) = \dim\mathcal{X} + \dim\mathcal{Y} - \dim(\mathcal{X}\cap\mathcal{Y}). \quad (4.4.19)$$

- length / number of vectors in any basis set

spanning set vs basis

1 any LI set

extend

2 basis set

3 any spanning set

$\{\alpha_1 v_1 + \alpha_2 v_2 + \cdots \alpha_i \cdots + \alpha_n v_n\}$

$= 0$ $= 2e$

suppose $\{u_1, \ldots, u_m\}$

$\in V$

LI

Proof Let $v_1, \ldots, v_m$ be a basis of $V_1 \cap V_2$; thus $\dim(V_1 \cap V_2) = m$. Because $v_1, \ldots, v_m$ is a basis of $V_1 \cap V_2$, it is linearly independent in V_1. Hence this list can be extended to a basis $v_1, \ldots, v_m, u_1, \ldots, u_j$ of V_1 (by 2.32). Thus $\dim V_1 = m + j$. Also extend $v_1, \ldots, v_m$ to a basis $v_1, \ldots, v_m, w_1, \ldots, w_k$ of V_2; thus $\dim V_2 = m + k$.

We will show that

2.44 $$v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k$$

is a basis of $V_1 + V_2$. This will complete the proof, because then we will have

$m + j + k =$

$(m + j) + (m + k) - m$

spans ✓

To prove that 2.44 is linearly independent, suppose

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0,$$

where all the a's, b's, and c's are scalars. We need to prove that all the a's, b's, and c's equal 0. The equation above can be rewritten as

2.45 $$c_1w_1 + \cdots + c_kw_k = -a_1v_1 - \cdots - a_mv_m - b_1u_1 - \cdots - b_ju_j,$$

which shows that $c_1w_1 + \cdots + c_kw_k \in V_1$. All the w's are in V_2, so this implies that $c_1w_1 + \cdots + c_kw_k \in V_1 \cap V_2$. Because $v_1, \ldots, v_m$ is a basis of $V_1 \cap V_2$, we have

$$c_1w_1 + \cdots + c_kw_k = d_1v_1 + \cdots + d_mv_m$$

for some scalars $d_1, \ldots, d_m$. But $v_1, \ldots, v_m, w_1, \ldots, w_k$ is linearly independent, so the last equation implies that all the c's (and d's) equal 0. Thus 2.45 becomes the equation

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0.$$

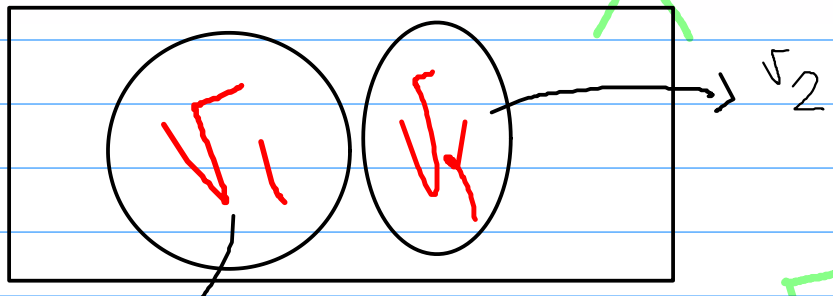
Because the list $v_1, \ldots, v_m, u_1, \ldots, u_j$ is linearly independent, this equation implies that all the a's and b's are 0, completing the proof. ∎

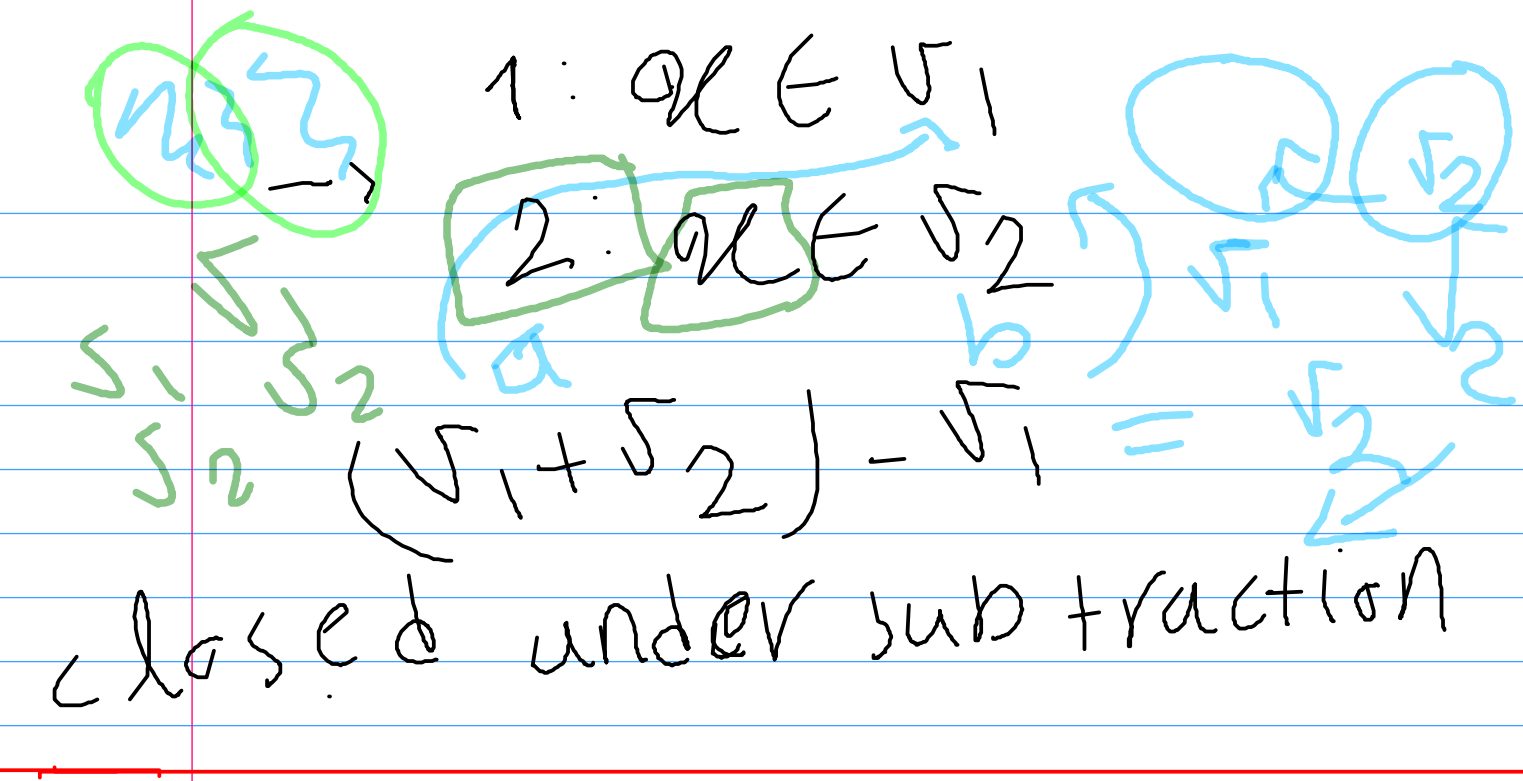
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11. Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be subspaces of a vector space $\mathcal{V}$ such that

$$\mathcal{V}_1 \cup \mathcal{V}_2$$

is also a subspace. Prove that one of the spaces is contained in the other one.





Cauchy–Bunyakovskii–Schwarz (CBS) Inequality

$$|\mathbf{x}^*\mathbf{y}| \le \|\mathbf{x}\|\,\|\mathbf{y}\| \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathcal{C}^{n\times 1}. \tag{5.1.3}$$

Equality holds if and only if $\mathbf{y} = \alpha\mathbf{x}$ for $\alpha = \mathbf{x}^*\mathbf{y}/\mathbf{x}^*\mathbf{x}$.

$x = y$

(1) $0 \le \|\alpha x - y\|$ $\quad T$

$0 \le \|\alpha x\|$

where $\alpha = \dfrac{x^T y}{\|x\|^2}$

$$0 \le \|x\|^2 \|y\|^2 - \|x^T y\|^2$$

$\|x\| = \langle x, x \rangle$

2

$$q = \frac{\langle x, y \rangle}{\|y\|^2} y + w$$

where $y \perp w$

$$\|q\|^2 = \frac{\|\langle x, y \rangle\|^2}{\|y\|^2} + \|w\|^2$$

$$\rightarrow \leq \frac{\|\langle x, y \rangle\|^2}{\|y\|^2}$$

$$q = cy + (x - cy)$$

$x = x + A - A$

$q = \alpha y + w$

scalar — where $y \perp w$

$\langle y, x - \langle y \rangle = 0$

$\rightarrow \langle y, x \rangle = \langle \|y\|^2$

$\rightarrow \langle = \frac{\langle y, x \rangle}{\|y\|^2}$

5.1.3. Show that $(\alpha_1 + \alpha_2 + \cdots + \alpha_n)^2 \leq n(\alpha_1^2 + \alpha_2^2 + \cdots + \alpha_n^2)$ for $\alpha_i \in \mathbb{R}$.

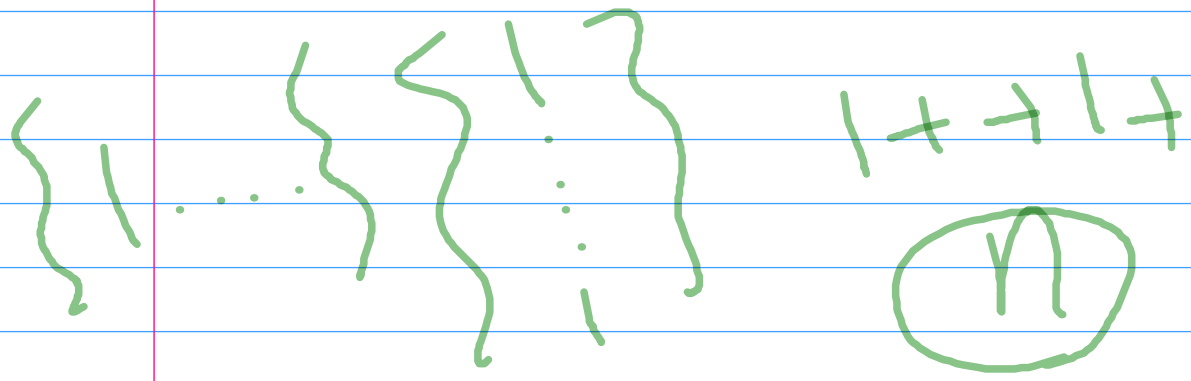
$x = \begin{Bmatrix} a_1 \\ \vdots \\ a_n \end{Bmatrix} \quad y = \begin{Bmatrix} 1 \\ \vdots \\ 1 \end{Bmatrix}$

$\| x^T y \| \leq \|x\| \|y\|$

LHS RHS

$x \downarrow \quad x^T \rightarrow \quad x^T y \quad \begin{Bmatrix} \alpha_1 \\ \alpha_n \end{Bmatrix}$ 1

$$\|x\|^2 = x^T x$$



I

$$\begin{Bmatrix} a_1 & \cdots & a_n \end{Bmatrix} \begin{Bmatrix} 1 \\ \vdots \\ 1 \end{Bmatrix} = \sum_{k=1}^{n} a_k$$

LHS

II

$$\|ze\|^2 = \|ze^T z\|$$

$$= \| \begin{Bmatrix} a_1 & \cdots & a_n \end{Bmatrix} \begin{Bmatrix} a_1 \\ \vdots \\ a_n \end{Bmatrix} \|$$

$$= a_1^2 + \cdots + a_n^2$$

RHS

LHS = RHS

6. Let $\boldsymbol{V} \in \mathbb{R}^{m \times n}$ be

$x_i \neq x_j$

$$\boldsymbol{V} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^{n-1} \end{bmatrix}$$

$m \times n$

where $x_i \neq x_j$, $\forall i \neq j$. Prove that the columns of $\boldsymbol{V}$ are linearly independent if $n \leq m$.

$\forall i \neq$

Vandermond matrice

$V\alpha = 0$ $\quad A\alpha = 0$

for each row $i =$

$$v_0 + \alpha_i v_1 + \cdots + \alpha_i^{n-1} v_{n-1} \quad \alpha \vec{e}$$

$$= p(\alpha_i) = 0 \qquad [\,,\,] = \vec{0}$$

$\longrightarrow$ n distinct roots /

FTA $\Longrightarrow$ $\deg(p) \leq n-1$ /

$n-1 < m$

$\rightarrow$ number of roots bigger than degree ?

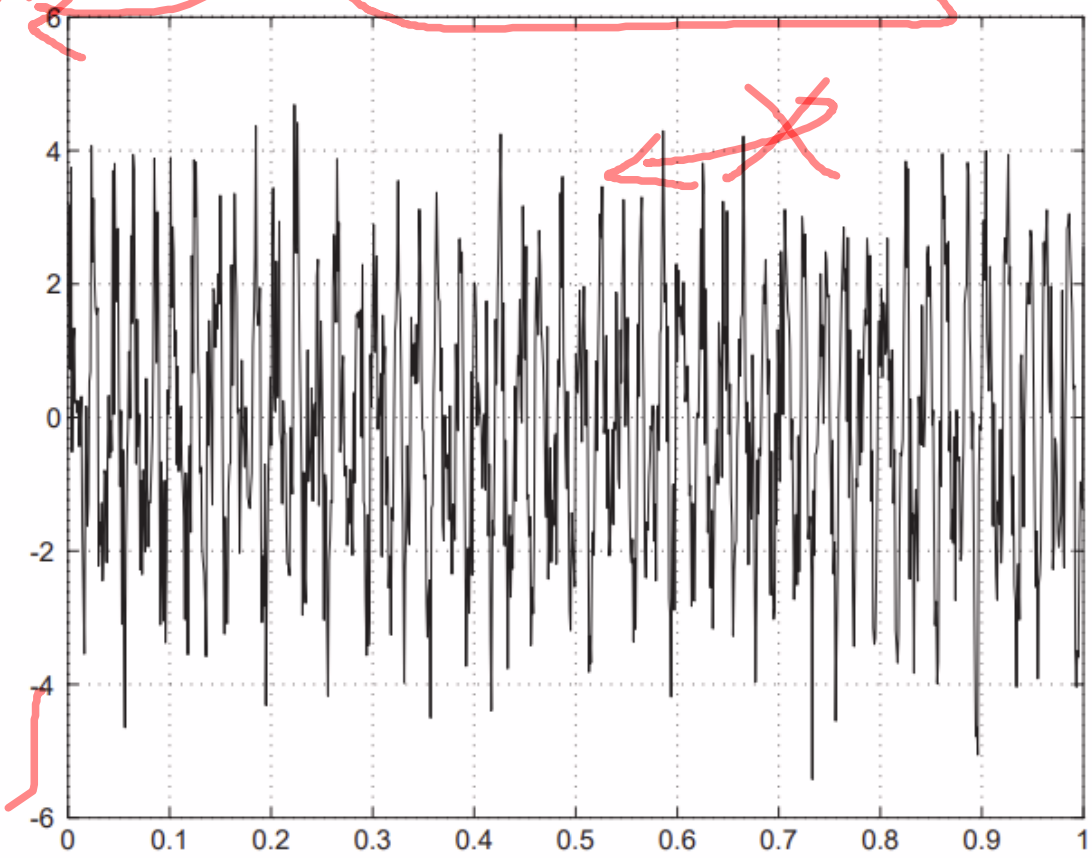
$\rightarrow$ only zero polynomial

$$\widehat{x}(\mathrm{e}^{\mathrm{j}\omega}) \triangleq \mathrm{DTFT}\{x[n]\}(\omega) \triangleq \sum_{m\in\mathbb{Z}} x[m]\,\mathrm{e}^{-\mathrm{j}\omega m} \triangleq \lim_{N\to\infty} \sum_{m=-N}^{N} x[m]\,\mathrm{e}^{-\mathrm{j}\omega m}$$

Fourier Matrix

The $n \times n$ matrix whose (j,k)-entry is $\xi^{jk} = \omega^{-jk}$ for $0 \le j,k \le n-1$ is called the ***Fourier matrix*** of order n, and it has the form

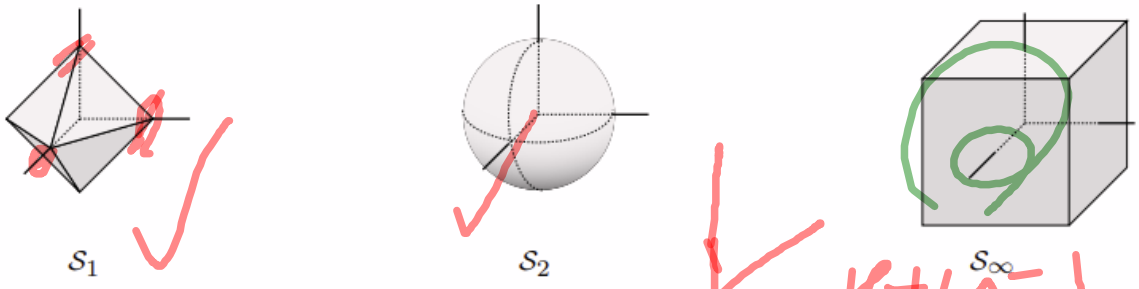
$$\mathbf{F}_n = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \xi & \xi^2 & \cdots & \xi^{n-1} \\ 1 & \xi^2 & \xi^4 & \cdots & \xi^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \xi^{n-1} & \xi^{n-2} & \cdots & \xi \end{pmatrix}_{n\times n}.$$



p-Norms

For $p \ge 1$, the ***p-norm*** of $\mathbf{x} \in \mathcal{C}^n$ is defined as $\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$.

To get a feel for the 1-, 2-, and ∞-norms, it helps to know the shapes and relative sizes of the ***unit p-spheres*** $\mathcal{S}_p = \{\mathbf{x} \mid \|\mathbf{x}\|_p = 1\}$ for $p = 1, 2, \infty$. As illustrated in Figure 5.1.4, the unit 1-, 2-, and ∞-spheres in $\Re^3$ are an octahedron, a ball, and a cube, respectively, and it's visually evident that $\mathcal{S}_1$ fits inside $\mathcal{S}_2$, which in turn fits inside $\mathcal{S}_\infty$. This means that $\|\mathbf{x}\|_1 \geq \|\mathbf{x}\|_2 \geq \|\mathbf{x}\|_\infty$ for all $\mathbf{x} \in \Re^3$. In general, this is true in $\Re^n$ (Exercise 5.1.8).



$\mathcal{S}_1$ $\mathcal{S}_2$ $\mathcal{S}_\infty$

$x+y=1$

$$\|x\|_1 = x_1 + \cdots + x_n$$

$$\Rightarrow \quad x+y+z=1$$

$$\|x\|_2 = \left(x_1^2 + \cdots + x_n^2\right)^{\frac{1}{2}}$$

$$\Rightarrow \quad \sqrt{x^2+y^2+z^2} = 1$$

$$\|x\|_\infty$$

$$= \max\{x_1, \cdots, x_n\}$$

why?

example:

$$x = \{1, 2, 3\}$$

$$\Rightarrow \|x\|_\infty = \lim_{p \to \infty} \left(1^p + 2^p + 3^p\right)^{\frac{1}{p}}$$

$x_1 \dots x_n$

$$= \lim_{p \to \infty} \left(3^p \left(1 + \frac{1^p}{3^p} + \frac{2^p}{3^p}\right)\right)^{\frac{1}{p}}$$

$$= \lim_{p \to \infty} 3 \left(1 + \left(\frac{1}{3}\right)^p + \left(\frac{2}{3}\right)^p\right)^{\frac{1}{p}}$$

$$= 3 = \max\{1, 2, 3\}$$

$\sum_{i=1}^{n} x_i^p$

$x_j > x_i \quad j \neq i$

$e^{p^{-1}}$

$\sqrt{1} = 1$

$2^{\frac{1}{p}} \to 2^0 = 1$

$3^{\frac{1}{p}} \to 3^0 = 1$

